
Applications of Large Language Models in Seismology: Probing Multimodal Reasoning on Waveform Interpretation Tasks

Advikar Ananthkumar

Harvard College

advikar_ananthkumar@college.harvard.edu

Moustafa Mousavi

Harvard University

Department of Earth & Planetary Sciences

Abstract

We investigate whether a multimodal large language model (LLM), specifically Gemini, can interpret three-component seismogram images and perform core seismological tasks including classification, phase picking, earthquake location, magnitude estimation, and first-motion polarity reading. Through a structured experimental progression from synthetic datasets to a curated 23-event real-earthquake catalog, we isolate the effects of prompt design, model scale, station geometry, and analyst-grade input alignment. We find that classification performance is highly sensitive to sampling and prompt context; location accuracy is dominated by station geometry rather than model capability, with a Geiger inversion using LLM-derived picks achieving a median epicentral error of 9.6 km (6.6 km after geometric quality control). The LLM produces magnitude estimates from raw-count seismograms with mean absolute error of 0.295 magnitude units (median 0.125 M) on the requested magnitude scale. In contrast, first-motion polarity reading remains at chance (47% agreement vs. analyst polarities) even when the input window is centered on the analyst’s published P-pick time, isolating the failure to a vision-model capability ceiling rather than a picking-error artifact. These results highlight both the promise and limitations of LLMs as scientific tools in seismology.

1 Introduction

Seismological analysis depends on the ability to interpret waveform data under noisy, incomplete, and time-sensitive conditions. When an earthquake occurs, analysts examine three-component seismograms to identify seismic phases, estimate arrival times, infer event magnitude, assess signal quality, and determine whether the event can be reliably located. These tasks are grounded in physics, but in practice they also require visual judgment: recognizing a weak P arrival, deciding whether a waveform is noise or signal, noticing whether station geometry is poor, and understanding when a source-parameter estimate should not be trusted. This expert labor is central to building and refining seismic catalogs, especially for events that are small, regional, poorly instrumented, or otherwise difficult to process automatically.

At the same time, multimodal large language models (LLMs) such as Gemini (2; 3), GPT-4V (4), Flamingo (5), and LLaVA (6) are increasingly being used as general-purpose scientific assistants (7). These systems can interpret images, follow domain-specific instructions, produce structured outputs, and combine visual evidence with textual reasoning. This raises a practical and scientific question:

can such a model perform parts of the interpretive work normally done by a seismologist, or does its performance break down once it is placed inside a realistic scientific workflow?

This paper investigates the following question:

Can a multimodal LLM interpret three-component seismogram images and carry out core pieces of a seismologist’s workflow?

We approach this question as a probing study rather than as a fixed benchmark. Our goal is not only to measure whether the model gets individual answers right, but to understand how an LLM-based pipeline holds up when used across the stages of seismic analysis. To do this, we move from controlled waveform datasets to a curated catalog of real earthquakes, and we evaluate the model on tasks including waveform classification, phase picking, magnitude estimation, first-motion polarity reading, and event location.

A central part of this work is the construction of a real-event catalog pipeline that mirrors the end-to-end workflow of a practicing seismologist: waveform retrieval, visual inspection, phase picking, source-parameter estimation, and location. This setting is important because LLMs are unlikely to be used as isolated classifiers in real scientific practice. Recent work places them as agents inside larger reasoning loops (8; 9) or as the orchestration layer for domain-specific scientific workflows (10; 11), where model outputs feed downstream algorithms and early errors propagate. Evaluating the model in this setting lets us ask not only whether Gemini can read a waveform image, but whether its outputs remain useful when connected to the rest of the seismological workflow.

This design lets us separate two kinds of difficulty that are often collapsed in simple accuracy metrics. Some tasks, such as event location, depend not only on the quality of the model’s visual interpretation, but also on station geometry, inversion stability, and the physical constraints of the seismic network. Other tasks, such as classification or approximate magnitude estimation, are more directly tied to what the model can infer from the waveform image itself. A central goal of the paper is therefore to distinguish failures of multimodal reasoning from failures caused by the structure of the underlying seismological problem.

This distinction is especially important for evaluating LLMs as scientific assistants. In a realistic workflow, an LLM does not merely answer isolated questions about an image; its outputs become inputs to later computations. A phase pick can affect an event location, a polarity estimate can affect a focal mechanism, and a magnitude estimate can shape how an event is interpreted. The pipeline developed here is designed to test whether multimodal LLM outputs remain useful once they are connected to these downstream scientific procedures.

This framing motivates two goals for the paper. First, we ask whether a multimodal LLM can reproduce parts of the visual reasoning that seismologists use when reading waveform records. Second, we ask whether those outputs remain useful when placed inside a larger analysis pipeline, where phase picks, magnitude estimates, and polarity readings become inputs to downstream location and source-characterization procedures. This distinction is important because scientific usefulness depends not only on whether a model can produce plausible interpretations, but also on whether those interpretations are stable enough to support later physical inference.

We first describe the waveform datasets and real-event catalog used in the study. We then introduce the LLM prompting and catalog-processing pipeline, including waveform retrieval, image generation, phase picking, magnitude estimation, event location, and focal-mechanism analysis. Finally, we evaluate performance across controlled experiments and real earthquakes, using the results to separate tasks where multimodal LLMs show practical promise from tasks where conventional seismological constraints dominate.

2 Data

2.1 Controlled waveform datasets

We first evaluate Gemini on controlled waveform datasets where labels are available and experimental conditions can be systematically varied. The primary dataset used in the early stages of the project is the Stanford Earthquake Dataset (STEAD) (12), which contains approximately 1.2 million three-component seismograms sampled at 100 Hz. We access STEAD through SeisBench (13) and use

paired waveform and metadata files, including earthquake traces from `chunk2.hdf5` and noise traces from `chunk1.hdf5`.

STEAD provides a useful starting point because it contains labeled three-component waveform records with associated metadata such as magnitude, depth, distance, and signal-to-noise ratio. However, it also introduces an important limitation: the STEAD earthquake traces used in this project are local and regional events. As a result, early experiments using STEAD alone artificially narrowed the classification problem and produced misleading behavior when the model was asked to distinguish local, regional, and teleseismic signals.

To broaden the experimental setting, we construct a merged dataset combining local/regional earthquakes from STEAD with additional teleseismic and non-earthquake sources. The merged dataset consists of STEAD earthquake traces, CREW teleseismic records, and Pacific Northwest records containing explosions and landslides. Before filtering, the merged dataset includes 150,000 STEAD samples, 5,071 CREW samples, and 5,000 PNW samples. We remove records with depth artifacts at 0 or 10 km and records falling outside the final bin definitions, producing a final merged dataset of 118,400 samples.

The merged dataset is stratified along four axes: distance, magnitude, depth, and signal-to-noise ratio (SNR). Distance is binned into $[0, 25)$ km, $[25, 100)$ km, $[100, 200)$ km, and $[200, \infty)$ km. Magnitude is binned into $[0, 3)$, $[3, 5.5)$, and $[5.5, \infty)$. Depth is binned into $[0, 40)$ km and $[40, \infty)$ km. SNR is binned into low-SNR and high-SNR regimes using a 30 dB threshold. This stratification is used to reduce sampling artifacts and to test whether model behavior changes across physically meaningful event regimes.

2.2 Real-event earthquake catalog

In addition to controlled waveform datasets, we construct a curated real-event earthquake catalog to evaluate the model in a more realistic seismological workflow. A summary of the catalog is provided in Table 1. The catalog contains 23 manually selected earthquakes ranging from M2.0 to M7.3 across California, Alaska, Washington, Nevada, Texas, Oklahoma, North Carolina, Mexico, and Turkey. The events are deliberately heterogeneous: the catalog includes small local events, larger regional events, offshore earthquakes with one-sided station coverage, sparse induced-seismicity events, and manually sourced Turkey events with limited waveform availability.

This design is intentional. Rather than selecting only clean, well-instrumented events, the catalog functions as a stress test for whether an LLM-based seismology pipeline can operate under realistic conditions. Each event is assigned a canonical identifier of the form `<REGION>_<YYYYMMDD>_M<XX>`, which links waveform files, station metadata, Gemini outputs, location results, and summary tables. Event-level metadata are taken from USGS when available, including latitude, longitude, magnitude, magnitude type, and event identifier. Two Turkey events are incorporated using AFAD waveform data because standard FDSN waveform access was unavailable.

Waveforms are retrieved from regional FDSN sources when available, including IRIS/EarthScope, SCEDC, NCEDC, TEXNET, KOERI, and EIDA routing. For each event, the downloader adaptively widens the search radius until enough stations are obtained, then stores waveform files and station metadata in an event-specific directory. These station-event waveforms form the input to the catalog pipeline described in the next section. Figure 1 shows the spatial distribution of the 23 catalog events.

3 Methods

3.1 Pipeline architecture

The catalog pipeline¹ (Figure 2) consists of eight stages, only two of which involve the LLM. Stage 1 selects events from the catalog and fetches USGS metadata. Stage 2 retrieves three-component waveforms from FDSN data centers. Stage 3 plots each station–event pair as a six-panel raw + bandpass image. Stage 4 (the first LLM call, called the seismogram-interpretation pass) emits a 19-field structured response per image: classification, phase picks, distance, depth, magnitude, polarity, and reasoning. Stage 5 (the second LLM call, called the polarity-reading pass) operates on a

¹Code, prompts, and result tables: <https://github.com/AdvikarA/Seismo-LLM> (1).

Event ID	Date	Region	Mag.	Type	USGS ID
TUR_20250810_M61	2025-08-10	Turkey	6.1	Mw	us6000qznb
TUR_20250423_M62	2025-04-23	Turkey	6.2	Mw	us7000pufs
CA_20250811_M21	2025-08-11	California	2.1	ML	nc75221857
CA_20250807_M21	2025-08-07	California	2.1	ML	ci41255944
CA_20250810_M24	2025-08-10	California	2.4	ML	nc75221282
CA_20250809_M20	2025-08-09	California	2.0	ML	nc75221217
AK_20250712_M44	2025-07-12	Alaska	4.4	ML	ak0258ve2yvm
AK_20250716_M73	2025-07-16	Alaska	7.3	Mw	us7000qd1y
CA_20250226_M33	2025-02-26	California	3.3	ML	nc75139226
OK_20240809_M32	2024-08-09	Oklahoma	3.2	ML	ok2024pnwx
NV_20241209_M57	2024-12-09	Nevada	5.7	Mw	nn00888580
TX_20250130_M46	2025-01-30	Texas	4.6	Mw	tx2025cbg1
CA_20250629_M45	2025-06-29	California	4.5	ML	nc75202001
CA_20250526_M46	2025-05-26	California	4.6	ML	nc75186621
NM_20250504_M54	2025-05-04	New Mexico	5.4	Mw	tx2025iqwk
CA_20250414_M52	2025-04-14	California	5.2	Mw	ci40925991
WA_20250303_M45	2025-03-03	Washington	4.5	ML	uw62078906
MX_20241014_M50	2024-10-14	Mexico	5.0	Mw	us6000nyjn
CA_20221220_M64	2022-12-20	California	6.4	Mw	nc73821036
CA_20230101_M54	2023-01-01	California	5.4	Mw	nc73827571
TX_20221216_M52	2022-12-16	Texas	5.2	Mw	tx2022yp1g
NC_20200809_M51	2020-08-09	North Carolina	5.1	Mw	se60324281
MX_20221122_M62	2022-11-22	Mexico	6.2	Mw	us7000irjd

Table 1: Curated real-event earthquake catalog used for the end-to-end pipeline. The catalog spans low-magnitude local events, larger regional events, offshore events, and sparse-network events

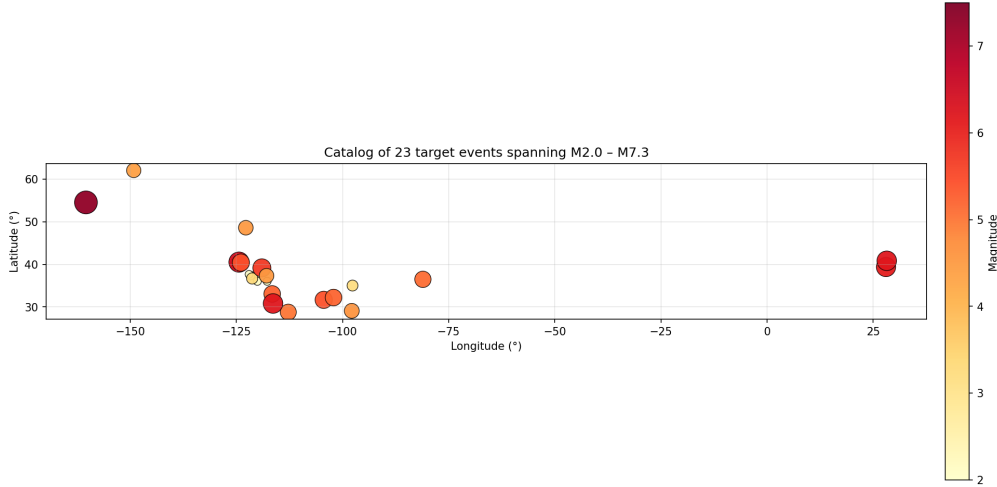


Figure 1: Real-event earthquake catalog. 23 events curated by hand to span M1.5–M7.3 across California, Alaska, Cascadia, Nevada, Texas, Oklahoma, North Carolina, Mexico, and Turkey. Markers are scaled by magnitude.

four-second high-pass-filtered zoom around the predicted P arrival, returning a sharper first-motion polarity and back-azimuth. Stage 6 inverts the seismogram-interpretation P-picks for hypocenter via Geiger’s method with Levenberg–Marquardt damping. Stage 7 inverts the polarity-reading polarities for a focal mechanism via a pure-Python HASH grid search. Stage 8 compares all outputs against USGS-published ground truth (event location, magnitude on the requested scale, analyst phase picks, analyst polarities, and analyst focal mechanisms where available).

The seismogram-interpretation and polarity-reading passes are the two LLM calls; everything else is deterministic Python that reads artifacts produced by an earlier stage. This separation lets us hold any subset of stages fixed while ablating another, which is the foundation of the five-run progression in

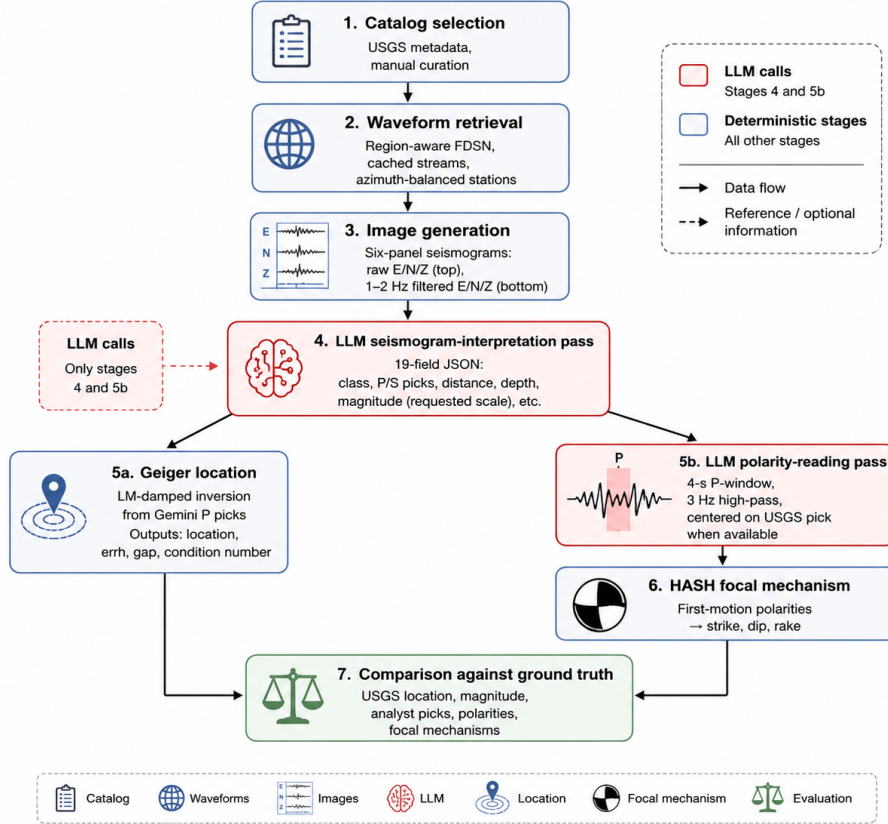


Figure 2: End-to-end real-event catalog pipeline. The red-highlighted stages are the only LLM calls: the full seismogram-interpretation pass and the polarity-reading pass. All other stages are deterministic, allowing model outputs to be evaluated both directly and after propagation through downstream seismological procedures.

Section 4. The remainder of this section describes each stage in detail, followed by the evaluation methodology used in Section 4.

3.2 Catalog selection and event identification

The 23 events used in the real-event evaluation (Table 1) were curated by hand to span a range of magnitude, network density, source mechanism, and data-source quality. Each event is assigned a canonical identifier of the form `<REGION>_<YYYYMMDD>_M<XX>`, which serves as the join key into every downstream artifact (waveform directories, station metadata, seismogram-interpretation and polarity-reading output rows, location results, summary tables). Event-level metadata are pulled from USGS via direct event-ID lookup, including epicenter coordinates, depth, reported magnitude, magnitude type (ML or Mw), and the USGS place description. Two Turkey events are incorporated using AFAD waveform data because standard FDSN waveform access was not available; these events flow through the same pipeline downstream of waveform retrieval.

3.3 FDSN waveform retrieval

Waveforms are pulled from FDSN data centers (15) using a region-aware, multi-source, parallel downloader implemented on top of ObsPy (14). For each event, candidate stations are first enumerated by querying station metadata within an initial search radius (5°), restricted to broadband and high-frequency channel codes (HH?, BH?, EH?). Each candidate is then submitted to a thread pool with up to eight concurrent workers. Each worker tries an ordered list of FDSN sources specific to the event’s region: NCEDC for Northern California events (with a direct-HTTP path to bypass IRIS-mirror

latency), SCEDC for Southern California, TEXNET for Texas and Oklahoma induced seismicity, KOERI for Turkey events, and the IRIS data management center plus EIDA routing for everything else.

If fewer than the target number of stations (eight) returns waveforms at the initial radius, the radius is widened through the schedule $\{5^\circ, 8^\circ, 12^\circ, 18^\circ, 25^\circ\}$ until the target is reached or the schedule is exhausted. Successfully downloaded waveforms are merged into one ObsPy `Stream` per station, written to disk as MiniSEED, and station metadata (network, station, distance, latitude, longitude, channel count) are appended to a per-event `station_coords.csv`.

3.4 Azimuth-balanced station selection

The default closest- N heuristic produces one-sided geometries for events near network edges, offshore events, and events with all stations clustered in one direction. Such configurations make the location inversion ill-conditioned: $G^T G$ becomes near-singular along the under-sampled azimuth. We mitigate this with a balanced station selector. For each event, candidate stations are grouped into eight angular bins of 45° each, sorted by distance within each bin, and then picked circular order so that the first eight selections span all eight bins when populated. The selections are submitted to the thread pool in this order, so the early-stop criterion preserves geometric balance. The selector tolerates events with empty bins (e.g. offshore events with no station to the west) by simply skipping empty bins in the round-robin pass.

3.5 Image generation

Each station–event pair is plotted as a single image consisting of six panels arranged in two rows (Figure 4). The top row shows the three components (E, N, Z) of the raw waveform, while the bottom row shows the same three components after a fourth-order zero-phase Butterworth bandpass at 1.0–2.0 Hz. The bandpass improves P-onset visibility without obscuring local high-frequency content. Time windows are computed from the USGS origin time and TauP-predicted (17) P/S arrival times under the IASP91 reference Earth model (18), so that each trace contains approximately 30 s of pre-event noise and extends beyond the predicted S arrival and coda. Sample rates are normalized to 100 Hz.

For comparison, Figure 3 shows only the raw three-component traces without the filtered row. The seismogram-interpretation pass receives the full six-panel representation because the filtered traces make phase arrivals visually easier to identify while preserving the original waveform context.

Several visual conventions were found to be important for reliable LLM interpretation. The horizontal axis is expressed in seconds since trace start rather than sample index; in earlier experiments, the model frequently failed to convert sample indices into physical time. Major tick spacing adapts to trace length, while minor ticks are placed every second with a faint secondary grid. This improved phase-pick precision without excessively cluttering the image. Figure dimensions, line widths, DPI, and color schemes are held fixed across all events to prevent the model from exploiting superficial visual differences between datasets or rendering styles.

A separate four-second waveform zoom is generated for the polarity-reading pass (Figure 5). This representation is centered on official P-wave arrival and high-pass filtered at 3 Hz to sharpen the first-motion onset, which is often partially obscured by low-frequency drift in the full waveform view.

3.6 LLM prompting

We use Gemini 3.1 Pro (gemini-3.1-pro-preview on Vertex AI, region `global`) for the primary results and Gemini 2.5 Flash-Lite for the smaller-model baselines, both members of the family described in (2; 3). All calls use sampling temperature 0.1 and a JSON-mode response schema; an exponential-backoff retry loop ($2 \rightarrow 4 \rightarrow 8 \rightarrow 16$ s, 5 retries) handles transient quota errors. The pipeline issues one seismogram-interpretation call per station–event pair (typically eight per event), and one polarity-reading call per surviving polarity candidate (capped at the closest 10 stations per event, restricted to events with $M \geq 3$ where first-motion is above the readability floor).

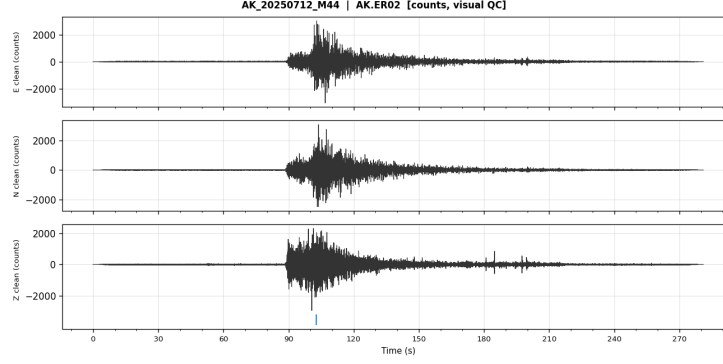


Figure 3: Three-component raw waveform from station AK.CUT for a M4.4 Susitna AK earthquake (canonical event AK_20250712_M44). E (east), N (north), Z (vertical) traces plotted against trace-start-relative time in seconds, with adaptive major ticks and 1-s minor grid. This is the simplest representation a seismologist consumes; for the LLM the equivalent input is the six-panel raw + bandpass version in Figure 4.

Seismogram-interpretation prompt. The seismogram-interpretation pass receives the six-panel image in Figure 4 and returns a structured JSON object containing 19 fields. These fields fall into six groups:

- **Waveform classification:** waveform_classification
- **Distance and depth classification:** earthquake_distance_classification, earthquake_depth_classification
- **Phase picks:** p_arrival_time_s, p_wave_type, s_arrival_time_s, s_wave_type
- **Polarity and wave indicators:** p_first_motion, has_lg_wave, has_love_wave, has_rayleigh_wave, has_depth_phases
- **Geometry and source estimates:** back_azimuth_degrees, incident_angle_degrees, epicentral_distance_km, depth_km, magnitude_m, magnitude_type, origin_time_wrt_p_s
- **Reasoning output:** reasoning (maximum 200 words)

The prompt embeds standard seismological background information, including wave types, distance regimes, depth phases, and magnitude-scaling conventions. It also enforces internal consistency constraints such as

$$p_{\text{arrival}} < s_{\text{arrival}}.$$

Prompt evolution. The prompt evolved through six revisions during the catalog phase. We briefly describe the three changes that affected results most. (i) Earlier prompts included an “anti-spike” rule that refused phase picks below 20 s into the trace, intended to suppress spurious early picks on noisy traces. The rule produced large regressions on legitimately-noisy events and was removed in favor of softer per-station distance- and class-consistency checks. (ii) The magnitude_type field was originally model-chosen, which produced cross-scale comparisons against USGS truth (e.g., predicted Mw vs. reported ML). We replaced it with a per-event substitution token {MAG_SCALE} populated from the catalog’s mag_type column, with explicit Hutton–Boore-style ML guidance (27) or Hanks–Kanamori-style Mw guidance (28; 29) inserted accordingly, so each prediction is on the same scale as the comparison ground truth. (iii) Gemini 3.1 Pro, on raw-count waveforms without instrument-response correction, returns magnitude_m: null on ~99% of stations citing missing calibration. We added a force-magnitude clause requiring a numeric magnitude estimate even when instrument response is unknown, with an instruction to use amplitude rules-of-thumb at the requested scale. This change recovers the magnitude metric without altering any other field.

Polarity-reading prompt. The polarity-reading pass receives the four-second zoom of Figure 5 and emits p_first_motion_z (UP / DOWN / UNKNOWN) along with a back-azimuth estimate. Where

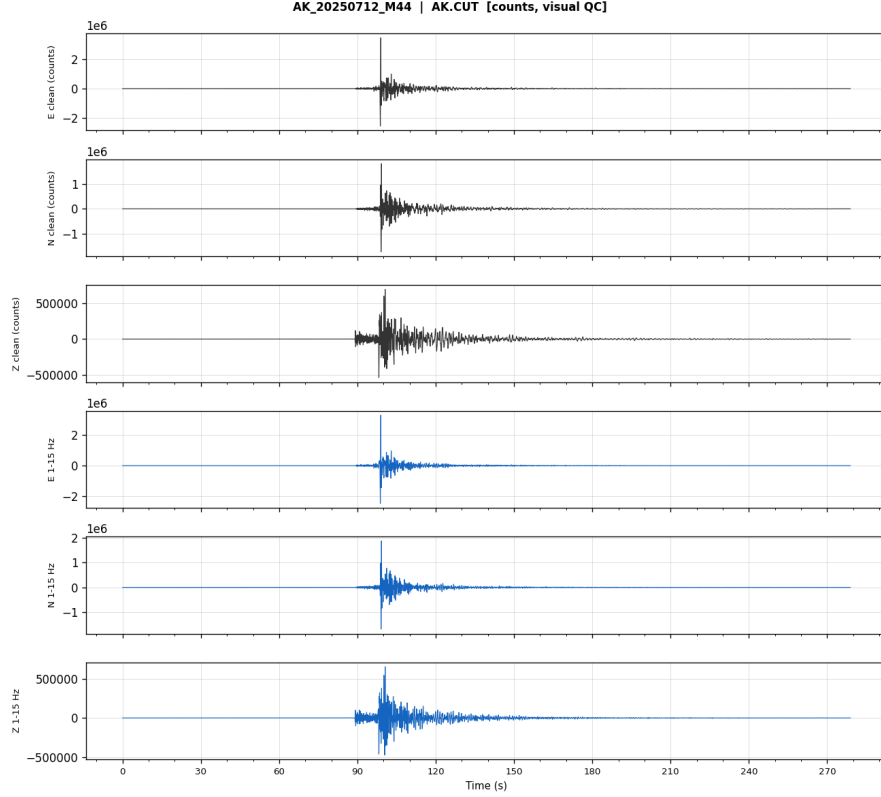


Figure 4: The six-panel image passed to the seismogram-interpretation pass for the same station–event pair as Figure 3. Top row: raw E, N, Z (counts). Bottom row: 1.0–2.0 Hz Butterworth bandpass of the same components. The bandpass row sharpens the P onset (around $t \approx 38$ s) without removing later high-frequency content needed for S identification. The model receives this entire image and emits a 19-field JSON response per call.

USGS phase-data is available for an event, the polarity-zoom window is centered on the analyst’s published P-pick time rather than the seismogram-interpretation pass’s own pick (Section 4.5); the prompt is otherwise identical. A `pick_source` field records whether the analyst-pick or the model-pick was used so the two cases can be analyzed separately.

3.7 Earthquake location

Locations are computed from the seismogram-interpretation pass’s P-wave arrivals using Geiger’s method (19; 20) with Levenberg–Marquardt damping (21). We implement this directly in Python rather than calling a conventional locator such as HYPOINVERSE-2000 (22) or HypoDD (23) so that the inverse problem is fully transparent inside the catalog notebook. The forward model evaluates P-wave travel times under a constant velocity $V_p = 6.0$ km/s with depth fixed to the USGS-published value (an option to solve for depth as a fourth unknown is available but not used in the primary results). At each iteration the Jacobian G of travel-time residuals with respect to $(\text{lat}, \text{lon}, t_0)$ is computed by central differences with step sizes of 0.01° for horizontal coordinates and 0.1 s for origin time. The update direction is the LM-damped least-squares solution

$$\Delta x = (G^T G + \lambda I)^{-1} G^T r,$$

where r is the residual vector and λ is the LM parameter. Each candidate step is clipped element-wise to a maximum of 0.5° in latitude and longitude and 30 s in origin time before being evaluated. If the new RMS residual is lower than the previous iterate, the step is accepted and λ is divided by 2; otherwise the step is rejected and λ is multiplied by 10. The iteration terminates when the horizontal step falls below 0.05 km or after 20 iterations.

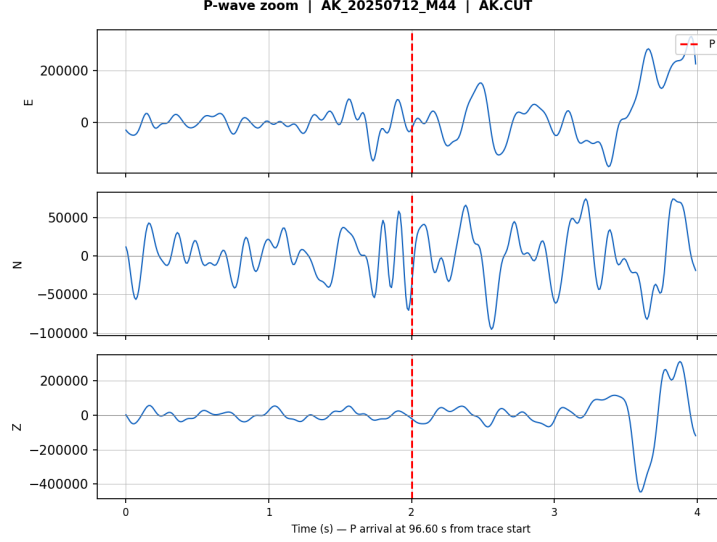


Figure 5: The four-second polarity-reading-pass input for the same station–event pair, centered on the analyst-published P-pick time and 3 Hz high-pass filtered. The high-pass removes a low-frequency drift that otherwise obscures the first-motion onset. The polarity-reading pass returns UP / DOWN / UNKNOWN per component along with a back-azimuth estimate.

For each located event we additionally report uncertainty diagnostics derived from the converged normal equations: errh_{km} (semi-major axis of the 95% horizontal confidence ellipse, computed by diagonalizing $(G^T G)^{-1}$ scaled by $\sigma^2 = \text{SSE}/(N - p)$), errt_s (origin-time 95% uncertainty), gap_{deg} (largest azimuthal gap between adjacent stations, sorted by bearing from the epicenter), $\text{cond}(G^T G)$ (condition number of the normal equations), iteration count, and a converged flag. These diagnostics are used in the quality-control filter described in Section 4.

3.8 Polarity estimation and focal mechanisms

For each event we fetch the USGS phase-data product, parse the QuakeML XML (16), and extract per-station P-wave picks (time and analyst-reported polarity, where present). The pick times center the polarity-reading zoom; the polarities serve as ground truth for the polarity-comparison evaluation. For events without USGS phase-data coverage (manual-data Turkey events, temporary network deployments) the polarity-reading zoom falls back to the seismogram-interpretation pass’s pick.

Focal mechanisms are computed from polarity-reading outputs using a pure-Python implementation of the HASH algorithm (24; 25); ground-truth comparison uses the USGS-published moment-tensor solution where available and the Global CMT catalog (26) as a secondary reference. For each candidate (strike, dip, rake) on a 5° grid, the radiation-pattern polarity at each station’s (takeoff, azimuth) position on the focal sphere is computed under a double-couple source assumption, and a quality score is the fraction of stations matching the model-emitted polarity. Solutions are visualized as Lambert equal-area lower-hemisphere stereographic projections (“beachballs”), with station polarities overlaid as filled (compression) or open (dilatation) circles. A solution is flagged as unreliable when the quality score falls below 0.7. Figure 6 shows the catalog’s best-aligned example, the M6.4 Ferndale California event, alongside the USGS-published moment-tensor solution for direct comparison.

3.9 Evaluation methodology

Per-station ground truth. For every station–event pair where USGS phase-data exists we extract the analyst-published P-pick time and convert to seconds-from-trace-start. Per-station signal-to-noise ratio is estimated as $20 \log_{10}(\text{RMS}_{\text{signal}}/\text{RMS}_{\text{noise}})$ in dB, with the noise window taken as the first 20 s of the trace (pre-event) and the signal window taken as the ± 2 s interval around the analyst pick. Polarity ground truth is the analyst-reported `Pick.polarity` field, mapped from positive/negative/undecidable to UP/DOWN/UNKNOWN.

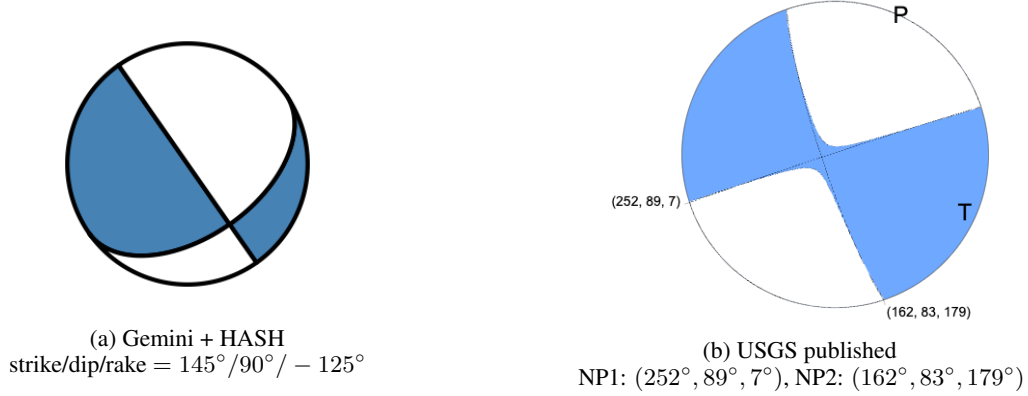


Figure 6: Focal-mechanism comparison for the M6.4 Ferndale California event (CA_20221220_M64), the catalog’s most accurate per-event polarity match (3 of 4 ground-truth-comparable stations agree; HASH quality 0.75). **(a)** Our HASH solution (lower-hemisphere Lambert beachball) inferred from the polarity-reading pass’s UP/DOWN reports. **(b)** USGS published moment-tensor solution with both nodal planes labeled. Although Ferndale is the catalog’s best event-level match on per-station polarity, the resulting focal-mechanism orientations differ substantially: the USGS strike-slip mechanism (vertical nodal planes near 162° and 252°) does not coincide with our oblique reverse solution. This illustrates the §4.5 finding that the underlying inverse problem is too poorly constrained for the resulting beachball to recover ground truth on a per-event basis.

Aggregate metrics. For phase picking, we report median and mean absolute residuals, Pearson r between predicted and reference times, fraction within $\{0.5, 1.0, 2.0\}$ s of analyst pick, and pick error stratified by SNR bin. For magnitude, we report mean absolute error, median absolute error, fraction within $\{0.5, 1.0\}$ magnitude unit of USGS, and per-event residuals on the requested magnitude scale. For location, we report great-circle distance between predicted and reference epicenters, the median and mean over the catalog, and the same metrics restricted to events that pass the geometric quality-control filter ($\text{gap}_{deg} < 250$ and $\text{cond}(G^T G) < 10^4$).

Polarity comparison. The polarity-comparison evaluation is performed in a separate post-hoc pass that joins the QuakeML-extracted analyst polarities with the polarity-reading outputs by $(\text{event_id}, \text{station_id})$. Only rows where both the analyst and the model committed to UP or DOWN are scored; UNKNOWN responses on either side are excluded. Agreement is reported pooled and stratified by pick_source (analyst-aligned vs. model-aligned input window) and by SNR bin.

3.10 Preceding STEAD-era methodological development

The catalog pipeline is the culmination methodological work on synthetic and pre-curated waveform datasets, primarily the Stanford Earthquake Dataset (STEAD). This phase was not a separate study but a sustained development effort that determined how the later real-event pipeline should be designed. It established five choices that carried directly into the catalog experiments: stratified sampling, standardized second-axis waveform images, unmarked inputs, structured JSON outputs, and SNR-aware evaluation.

The earliest STEAD pipeline established the core waveform-to-LLM scaffold. We loaded STEAD chunk2 from Google Cloud Storage, filtered traces by signal-to-noise ratio, generated three-component waveform plots, and queried Gemini 1.5 Pro through Vertex AI at temperature 0.1.

The same development phase revealed that visual representation strongly affected model behavior. Early waveform plots used sample index rather than physical time, and the model often failed to translate this axis into meaningful arrival-time estimates. Later plots therefore used seconds on the horizontal axis, fixed tick spacing, and standardized figure styling. We also tested versions with and without overlaid P/S arrival markers. Marker lines improved phase-pick-dependent metrics but leaked ground truth into the input image; the catalog pipeline therefore uses unmarked six-panel waveform images for the main seismogram-interpretation pass. We also moved from free-form text outputs

parsed by regex to a JSON-schema response, which made the LLM output directly consumable by the downstream deterministic location and focal-mechanism stages.

A central practical bottleneck during this phase was throughput. Running a sufficiently capable model (Gemini 2.5 Pro or its predecessors) on the full 118,400-trace stratified sample at the per-call latency measured on a personal machine would have taken roughly nine days of continuous inference, which was infeasible to operate as part of an interactive development loop. We collaborated Google to push the run through. The collaboration produced a two-tier prompting strategy and a corresponding execution path. We use the more capable model only to *generate the system prompt*, by feeding it labeled examples and asking it to produce a reasoning recipe and a corresponding system prompt for a faster model. The lower-latency model (Gemini 2.5 Flash, and later Gemini 2.5 Flash-Lite) then handles the bulk inference. The Google team then ran the full 118,400-trace stratified sample.

The STEAD-era experiments also established evaluation patterns the catalog phase inherits. Earthquake-versus-noise classification was strong in a balanced zero-shot setting with Flash, suggesting that basic classification was not the limiting task. Absolute distance estimation remained unreliable, with predictions often collapsing to round values such as 80, 5,000, or 6,000 km regardless of true distance. SNR-stratified analysis showed that low-SNR traces produced worse phase-pick behavior, motivating the later use of SNR as a quality-control variable in the catalog pipeline (Section 3.9).

3.11 Public-facing AI-generated catalog interface

To make the pipeline outputs inspectable by readers without re-running the inference, we built a public-facing catalog website that mimics the visual language of the official USGS earthquake event pages but is populated entirely from our LLM-derived outputs (Figure 7). Each catalog event is rendered as an event page that shows the per-station six-panel waveform images, the seismogram-interpretation pass’s reasoning string and structured-output table, the polarity-reading-pass beachball, the Geiger location with its uncertainty diagnostics, and a side-by-side comparison against the matched USGS reference page.

The site is intentionally framed as an AI-generated counterpart to the USGS catalog. The interface is generated as static HTML from the same `master_summary.csv`, `step1_gemini.csv`, and per-event waveform/zoom/beachball PNGs that drive the paper’s tables and figures.

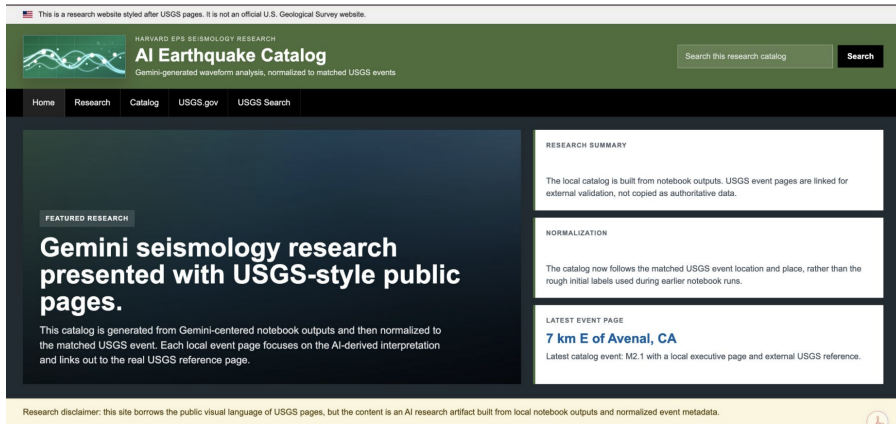


Figure 7: Landing page of the AI-generated catalog website. The visual language deliberately mirrors USGS event pages, but all per-event content (waveforms, picks, location, magnitude, polarity, focal mechanism) is sourced from the LLM-driven catalog pipeline rather than from analyst data.

4 Experiments

We organize the catalog evaluation as a five-run controlled progression, each run changing one component of the pipeline while holding the rest fixed. The progression is followed by per-task

Run	Configuration	Med. Loc	Mean Loc	Mag MAE	Med. Mag	Null %
1	Flash-Lite, no rules	16.4 km	–	–	–	–
2	Flash-Lite + rules	15.4 km	47.4 km	1.50	1.47	41%
3	3.1 Pro, null allowed	11.3 km	50.5 km	–	–	99%
4	3.1 Pro + force-mag, simplex	11.4 km	55.2 km	0.42	0.27	0%
5	3.1 Pro + force-mag + Geiger + AB + USGS-zoom	9.6 km	59.2 km	0.295	0.125	0%

Table 2: Run progression showing the effect of prompt rules, model scale, forced magnitude estimation, Geiger inversion, azimuth-balanced (AB) station selection, and USGS-pick-centered polarity zoom. Runs 1–3 use a Nelder–Mead simplex locator; Runs 4–5 use Geiger’s method with Levenberg–Marquardt damping. Run 5 uses 23 events; the earlier runs use the prior 22-event catalog.

analyses (magnitude, phase picking, location with quality control, and the polarity ablation) that report the Run 5 numbers in detail.

4.1 Five-run progression

We evaluate the behavior of the proposed pipeline through a sequence of five controlled runs. Each run modifies one identifiable component (model, prompt, locator, station selector, or zoom-window centering) while holding the rest of the pipeline fixed, and is executed end-to-end on the same 22-event catalog (Run 5 additionally includes one event added to the catalog after Run 4). All runs use sampling temperature 0.1, the same six-panel image generation, and the same FDSN cache when not explicitly invalidated. Results are summarized in Table 2.

Run 1: baseline. Gemini 2.5 Flash-Lite with a minimal prompt that simply elicits the structured output schema, no consistency rules, no force-magnitude clause. Locations are computed by Nelder–Mead simplex minimization on travel-time residuals. This serves as the lower bound that subsequent improvements are measured against.

Run 2: prompt rules. Same model and locator as Run 1, but the prompt now adds soft consistency rules: per-station epicentral distance must be consistent with the predicted phase pair, and the distance-class field must be consistent with the numeric distance estimate. We also ablated an aggressive “anti-spike” rule that refused phase picks below 20 s into the trace; that rule was found to make a Turkey event ten times worse (32 km $\rightarrow$ 333 km) by suppressing legitimate noisy picks, and was removed from this run forward.

Run 3: model upgrade. Switch from Gemini 2.5 Flash-Lite to Gemini 3.1 Pro, holding the prompt and locator fixed. Magnitude is permitted to be null. The 3.1 Pro model declines to estimate magnitude on $\sim 99\%$ of stations, citing the absence of instrument-response correction; the magnitude metric is therefore not informative for this run. This run isolates the effect of model capability on tasks the larger model is willing to commit to (classification, picking, location).

Run 4: force-magnitude. Same as Run 3, with two changes. (i) The prompt is augmented with a force-magnitude clause requiring a numeric `magnitude_m` estimate even when instrument response is unknown, with explicit amplitude rules-of-thumb at the requested magnitude scale. (ii) The simplex locator is replaced by Geiger’s method with Levenberg–Marquardt damping; this change rescues two events that had previously failed under simplex (TX_20250504: 364 km $\rightarrow$ 4 km; CA_20250526: 410 km $\rightarrow$ 41 km) by stabilizing the iteration when picks are noisy. Run 4 establishes the baseline for the magnitude metric (mean MAE 0.42 M, median 0.27 M).

Run 5: deterministic infrastructure. Same model and force-magnitude prompt as Run 4, with three additional methodological changes. (i) The FDSN station selector switches from closest-eight to azimuth-balanced (eight 45° bins, round-robin), and the cache is invalidated so a fresh pull populates the new selection. (ii) The polarity-reading-pass zoom window is centered on the USGS analyst’s published P-pick time where available (with a 3 Hz high-pass filter), and the seismogram-interpretation pass’s pick where not. (iii) The seismogram-interpretation prompt’s `magnitude_type` field is replaced with a per-event {MAG_SCALE} substitution token populated from the catalog (10

events ML, 13 events Mw), so each magnitude prediction is on the same scale as the comparison ground truth.

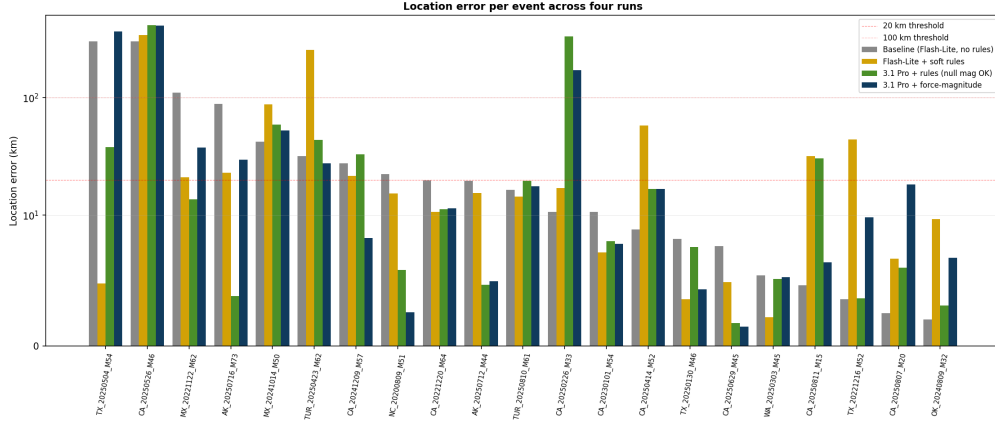


Figure 8: Per-event location error across Runs 2–5. The model-capability jump (Run 2 $\rightarrow$ 3) tightens the median; the locator change and force-magnitude prompt (Run 3 $\rightarrow$ 4) re-stabilize two previously failing events; the Run 5 azimuth-balanced selection produces the lowest median but introduces three regressions on events whose closest-station clusters were broken up to gain azimuthal coverage.

Run 5 wins the median on every metric. Its mean location error is slightly worse than Run 4 because three events regress sharply under the azimuth-balanced station selection (CA_20221220_M64: 10 $\rightarrow$ 451 km; CA_20250526_M46: 41 $\rightarrow$ 269 km; AK_20250716_M73: 32 $\rightarrow$ 118 km), in each case because balancing forced the inversion to use distant low-quality stations in place of clustered close-in ones. All three are correctly self-flagged by the Geiger uncertainty diagnostics (azimuthal gap $> 230^\circ$, condition number $> 10^3$), and after applying the quality-control filter introduced in Section 4.4 the median location error drops to 6.6 km on the 16 trustworthy events.

4.2 Magnitude estimation

Magnitude estimation is evaluated on events where the model produces a numerical output, with each prediction compared against USGS on the requested magnitude scale (ML or Mw) via a per-event scale substitution in the prompt. In Run 5, with magnitude output forced and the scale substitution active, the model achieves a mean absolute error of 0.295 magnitude units and a median error of 0.125 magnitude units across 23 events. Sixteen of the 23 events fall within 0.5 magnitude units of the USGS reference value, and 21 of 23 within 1.0 unit. Thirteen events are within 0.1 magnitude units. The single largest residual is 1.26 M on a Turkey event using AFAD waveform data with non-standard station calibration.

Compared to Run 2 (Flash-Lite with rules), which achieves a mean absolute error of 1.50 magnitude units, this represents approximately a $5\times$ reduction in mean error and over an order of magnitude reduction in median error. The improvement decomposes into two roughly equal components: a model-capability component (Run 2 to Run 4) reducing mean MAE from 1.50 to 0.42 magnitude units, and a methodological component (Run 4 to Run 5) reducing it further from 0.42 to 0.295. The latter is attributable to the per-event magnitude-scale substitution, which removes a cross-scale confound from the comparison metric. This entire result occurs despite the absence of instrument-response correction in the input waveforms. Figure 9 shows the per-event predicted-vs-truth scatter.

4.3 Phase-pick accuracy

We evaluate P-wave arrival-time predictions against analyst picks from USGS phase data. Across 117 matched station-event pairs from Run 5, the model achieves a median absolute residual of 0.56 seconds and a Pearson correlation coefficient of $r = 0.72$ between predicted and reference arrival times. Approximately 46% of picks fall within 0.5 s of the analyst pick, 67% within 1.0 s, and 85% within 2.0 s. The mean absolute residual is 3.0 s, but is heavy-tailed: a small number of low-SNR outliers dominate the mean.

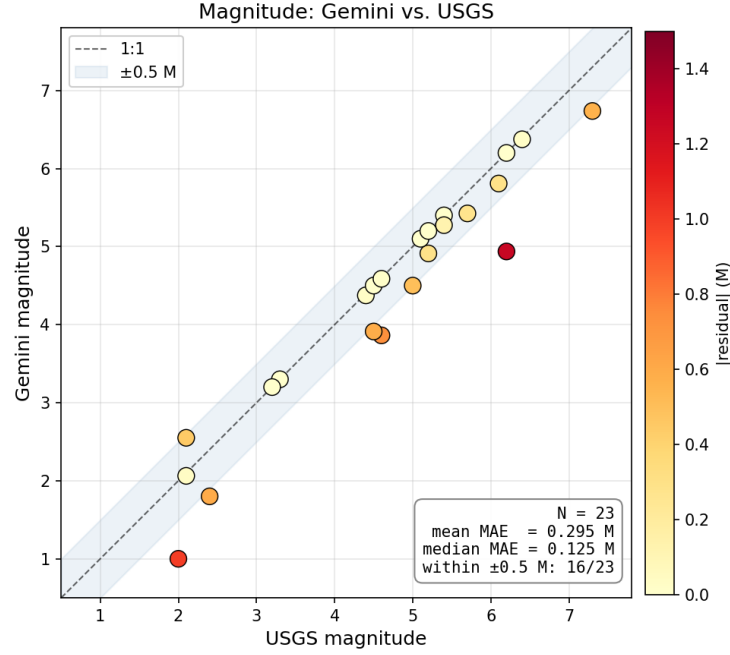


Figure 9: Run 5 magnitude estimates vs. USGS truth on the requested magnitude scale. Mean MAE 0.295 M, median 0.125 M, r to 1:1 line tight across two orders of magnitude in moment release. The single visible outlier is TUR_20250423_M62, served from AFAD waveforms with non-standard station calibration.

Performance is strongly dependent on signal quality. The median absolute residual stratified by SNR is 0.52 s for traces with $\text{SNR} \geq 30$ dB ($N = 99$), 0.52 s for 20–30 dB ($N = 9$), and 18.6 s for 10–20 dB ($N = 8$). Below approximately 20 dB the model fails to reliably identify P-wave arrivals; above 20 dB the model is at the level of analyst noise, supporting a deployment recommendation that LLM-based picks are usable only above the 20 dB SNR threshold. Figure 10 shows the per-station scatter, and Figure 11 shows the SNR-stratified residual.

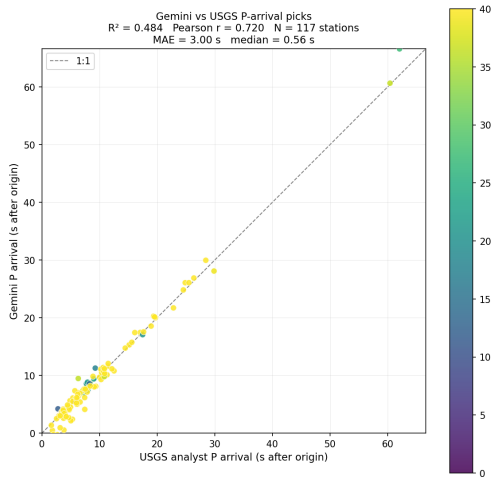


Figure 10: Predicted vs. analyst P-arrival time across 117 matched station–event pairs. Pearson $r = 0.72$; median $|residual| = 0.56$ s.

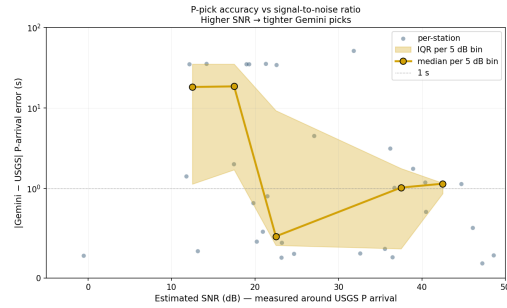


Figure 11: Pick error stratified by SNR. The cliff between 10–20 dB (~ 18 s) and ≥ 20 dB (~ 0.5 s) supports an SNR-thresholded deployment policy.

4.4 Earthquake location and geometric quality control

Run 5 produces a median location error of 9.6 km across 21 located events (the remaining 2 of 23 events fail at the data-availability stage with only one station returned, on which a location cannot be computed). Eleven events have errors below 10 km, 14 below 20 km, and 4 above 100 km. The Geiger inversion additionally yields per-event uncertainty diagnostics: azimuthal gap, condition number of the normal equations, the 95% horizontal error ellipse errh_{km} , and origin-time uncertainty errt_s . The median azimuthal gap is 142° (an improvement from 199° in Run 4 attributable to the azimuth-balanced station selector), with 9 of 21 events at gap $> 180^\circ$ and 5 of 21 at gap $> 250^\circ$. Median 95% horizontal error ellipse is 15.5 km, median origin-time uncertainty 1.20 s, median condition number 170, and median RMS travel-time residual 0.94 s.

The four events with location errors above 100 km all have azimuthal gaps exceeding 230° and condition numbers exceeding 10^3 , indicating that the underlying inverse problem is geometrically under-constrained on those events. This is most pronounced for an offshore event (CA_20250526_M46) where no stations exist in the eastern hemisphere relative to the epicenter, producing a condition number of 2.4×10^5 and a horizontal error ellipse of nearly 2,000 km. Applying a quality-control filter that retains only events with gap $< 250^\circ$ and condition number $< 10^4$ reduces the catalog to 16 events, on which the median location error drops to 6.6 km and the mean to 20.1 km. These diagnostics are therefore not only descriptive but operationally useful: they identify which event-level locations should be trusted before any comparison to ground truth.

4.5 Polarity ablation: window placement vs. vision-model capability

A central methodological question for first-motion polarity reading is whether observed model failures are caused by misalignment of the input window (a picking-error artifact) or by an inability of the vision model to resolve the polarity itself. Because earlier runs centered the polarity zoom on the LLM’s own pick, the two failure modes were entangled: a low polarity-agreement number could be explained either by the model misreading first motion or by the input window simply not containing the true onset.

We separate these two failure modes in Run 5 with a deliberate ablation. For every station–event pair covered by ANSS phase data, we fetch the USGS analyst’s published P-pick time from the phase-data product and use it (instead of the seismogram-interpretation pass’s own pick) as the center of the 4-second polarity-zoom window passed to the polarity-reading pass. A 3 Hz high-pass filter is applied to sharpen the first-motion onset. Stations not covered by ANSS phase data fall back to the seismogram-interpretation pass’s pick. Each output row records its centering source via a `pick_source` field so the two cases can be sliced post-hoc.

To score polarity agreement we then run an independent QuakeML pass over the same phase-data product, extracting each `Pick.polarity` value (`positive` / `negative` / `undecidable`) and mapping to UP / DOWN / UNKNOWN. Polarities are joined with the polarity-reading outputs by (`event_id`, `station_id`). Only rows where both the analyst and the model committed to UP or DOWN are scored; UNKNOWN responses on either side are excluded. Of the 51 resulting matched pairs, 9 used the analyst-aligned window and 42 used the model-aligned or pre-existing window. Results are summarized in Table 3.

Slice	<i>N</i>	Agreement
All committed UP/DOWN	51	47.1%
Window centered on USGS analyst P pick	9	33.3%
Window centered on LLM pick / pre-existing	42	50.0%
SNR ≥ 30 dB	38	55.3%
SNR 20–30 dB	1	–
SNR < 20 dB	3	0%

Table 3: First-motion polarity agreement between Gemini and the USGS analyst. The 9-row analyst-aligned slice mechanically removes picking-error from the input window placement; the model still fails to read polarity above chance.

The pooled agreement is 47.1%, statistically indistinguishable from chance under a binomial null. The SNR ≥ 30 dB slice reaches only 55.3%, again only marginally above chance. Most importantly,

the 9-row subset where the input window was provably centered on the analyst’s P-pick time does not show improved agreement; if anything the point estimate is lower, though the 95% confidence interval ($\sim \pm 30$ percentage points) is too wide to support a directional claim. The confusion matrix is approximately symmetric, with no systematic flip that could be corrected by relabelling. The model emits UP roughly 55% of the time regardless of the analyst-reported polarity.

This ablation rules out two natural hypotheses about the polarity failure: (i) that the model could read polarity if the window were positioned more accurately, since window placement was experimentally fixed at the analyst’s pick; and (ii) that the model could read polarity if the high-frequency content were boosted, since a 3 Hz high-pass filter was applied. Neither modification moved agreement above chance. The remaining hypothesis consistent with the data is that polarity reading is constrained by the vision model’s capability to resolve subtle first-motion features at the SNR and magnitude regime sampled by the catalog.

The downstream consequence is visible in Figure 12, which compares our HASH-derived focal mechanisms against published USGS / GCMT moment-tensor solutions for five catalog events spanning M3.2 to M6.4 across five regions and three fault types (strike-slip, normal, reverse / thrust). Our solutions and the reference solutions disagree systematically: where USGS reports near-vertical strike-slip nodal planes (Ferndale CA), our solution is oblique reverse; where USGS reports a near-horizontal normal mechanism (Bigadiç Turkey), our solution recovers the dip but rotates the strike; the small-magnitude Oklahoma and Orcas Island events are essentially unrelated. This is the visual companion to Table 3: even at the catalog’s best per-event polarity agreement (Ferndale, 75%), the resulting focal-mechanism orientation is not recovered, and the systematic mismatch across the five events is consistent with HASH operating on chance-level polarities rather than with HASH itself being miscalibrated.

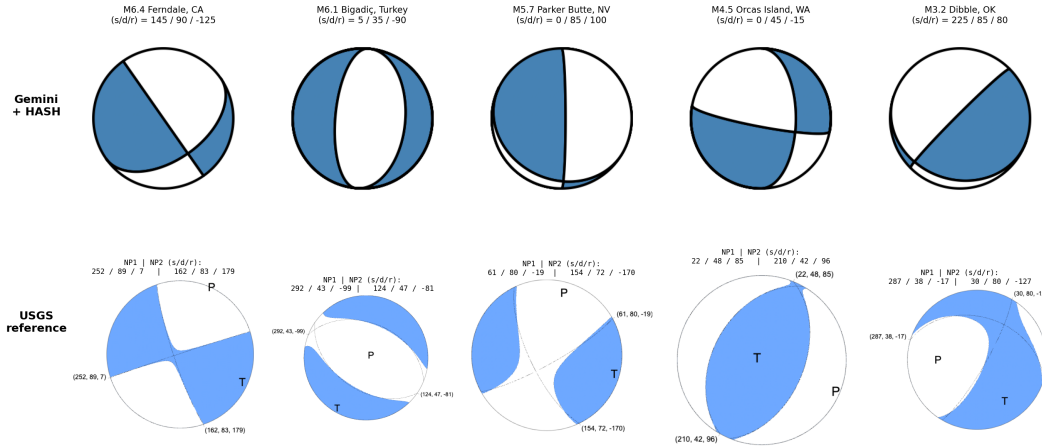


Figure 12: Catalog-wide beachball comparison across five events spanning M3.2–M6.4. Left column: our HASH solutions inferred from the polarity-reading pass. Right column: USGS-published moment-tensor / first-motion solutions. The five events together cover three fault types and five regions; in no case does our solution recover the reference orientation, consistent with the chance-level per-station polarity agreement reported in Table 3

5 Results

5.1 Separation of error sources

A central result of this study is that model performance separates along two distinct axes, corresponding to different sources of error.

Location accuracy. Errors in earthquake location are dominated by station geometry and inversion stability. Improvements in prompt design (Run 1 to Run 2) produce only marginal gains in median location error, while increasing model capability (Run 2 to Run 3) yields a more substantial reduction. Forcing the model to emit a magnitude (Run 3 to Run 4) does not improve location accuracy,

indicating that location performance is not bottlenecked by output completeness either. The largest single-step improvement in median location error comes instead from the deterministic infrastructure changes in Run 5 (Geiger inversion with Levenberg–Marquardt damping plus the azimuth-balanced station selector), which lower the median to 9.6 km, and the geometric quality-control filter further reduces this to 6.6 km on the trustworthy subset.

Inspection of individual events shows that large location errors are uniformly associated with high azimuthal gaps and ill-conditioned inverse problems. In these cases, even accurate phase picks cannot recover a stable solution, and the resulting errors are best understood as geometry-limited rather than model-limited. Critically, the Geiger inversion’s per-event uncertainty diagnostics flag these failure cases without reference to ground truth, supporting an operational deployment in which untrustworthy locations can be identified algorithmically.

Magnitude accuracy. In contrast, magnitude estimation is strongly dependent on model behavior and prompt design. Smaller models tend to produce approximate magnitude estimates even when uncertain, leading to relatively large errors. More capable models exhibit the opposite behavior: when allowed to abstain, they frequently decline to produce a magnitude estimate due to the absence of instrument-response information. When forced to produce a numerical estimate, however, the same model achieves substantially improved accuracy (mean MAE 0.295, median 0.125 in Run 5). This indicates that magnitude estimation performance is governed not only by model capability but also by output constraints imposed during prompting and by removing cross-scale confounds in the comparison metric.

Polarity accuracy. First-motion polarity reading is the only major task in our evaluation that did not improve across runs. The polarity ablation in Section 4.5 mechanically removes the picking-error confound by centering the input window on the analyst’s published P-pick time, and shows that polarity agreement remains at chance ($\sim 47\%$) even on this subset. Unlike the location and magnitude axes, where the model’s behavior is sensitive to surrounding methodology, polarity behavior appears insensitive to both window placement and high-pass filtering. We therefore interpret polarity as a vision-model capability ceiling rather than an artifact of the surrounding pipeline.

6 Discussion

6.1 Key positive finding

The most notable positive result of this study is that a state-of-the-art multimodal LLM, when required to produce a numerical estimate, can infer earthquake magnitude from raw-count waveform images with sub-quarter-magnitude median accuracy and sub-half-magnitude mean accuracy on a 23-event multi-region catalog. This occurs despite the absence of instrument-response correction, which is typically considered essential for magnitude estimation. The result holds when each estimate is compared on the requested magnitude scale (ML or Mw) on a per-event basis, removing cross-scale confounds.

This suggests that the model is able to extract amplitude-, duration-, and waveform-shape-related cues from images that correlate with event magnitude, even without access to physically calibrated ground-motion measurements. The accuracy is sufficient to be useful as a first-pass magnitude estimate in triage workflows, particularly when paired with a forced-output prompting strategy that prevents the model from abstaining under uncertainty.

6.2 Key negative finding: polarity reading is a vision-model ceiling

The most informative negative result is that first-motion polarity reading remains at chance even when the input window is centered on the USGS analyst’s published P-pick time. Earlier runs in the development of this pipeline left open the possibility that polarity failures were caused by misaligned input windows: the model was being asked to identify first motion at a centered position that did not in fact correspond to the true first arrival. The Run 5 ablation removes this confound mechanically by centering the polarity zoom on the analyst’s pick (Section 4.5), and the model still fails. The ablation rules out window placement and high-pass filtering as explanations, leaving the underlying limit at the vision model’s capacity to resolve subtle waveform features at the SNR and magnitude regime represented in the catalog.

This has direct downstream consequences. First-motion polarities are the input to focal-mechanism estimation via the HASH algorithm. With agreement at 47%, the polarities entering HASH are essentially uncorrelated with ground truth, so the resulting beachballs match analyst solutions only by accident. We therefore do not present per-event focal-mechanism accuracy as a primary result; we report the polarity finding instead, since it is what the data actually supports.

7 Limitations

Several limitations bound the conclusions of this study.

Catalog scale. The real-event evaluation uses 23 manually curated events, spanning M1.5–M7.3 across nine regions. The catalog is heterogeneous but not exhaustive: it contains no teleseismic events, only one event above M7, and is weighted toward local-to-regional events in the western United States. The polarity comparison is also limited by analyst coverage, with only 51 committed UP/DOWN station–event pairs available for scoring.

Methodological assumptions. The Geiger inversion assumes a constant P-wave velocity of 6.0 km/s and fixes depth to the USGS value, rather than using an event-specific velocity model. The HASH focal-mechanism stage uses a simplified first-motion grid search, while USGS mechanisms are often derived from richer waveform inversions. We also do not fully ablate choices such as the closest-10 station cap, the $M \geq 3$ polarity filter, or the exact force-magnitude prompt wording.

Model specificity. All primary catalog results use Gemini 3.1 Pro at temperature 0.1. The model remains non-deterministic, with observed run-to-run drift of roughly ± 5 –20 km in event location and ± 0.1 –0.2 M in magnitude. We do not evaluate other multimodal LLMs, so the reported results should be interpreted as Gemini-specific.

Ground truth. USGS phase picks, polarities, magnitudes, and mechanisms are treated as reference values, but they are themselves analyst products with uncertainty, especially for small or sparsely recorded events. The polarity analysis excludes undecidable picks, which may introduce a station-quality bias.

Reproducibility cost. Re-running the full catalog requires several hundred LLM API calls plus waveform retrieval from external data centers. The pipeline is reproducible from the released notebook and cache logic, but regeneration is not cost-free.

8 Conclusion

This paper presents a systematic empirical study of multimodal LLM performance on seismological waveform interpretation tasks. By embedding the model within an end-to-end catalog pipeline that mirrors the workflow of a practicing seismologist, we evaluate not only its ability to interpret waveform images, but also the reliability of its outputs when used in downstream scientific analysis.

Across a five-run progression on a 23-event real-earthquake catalog, we find that LLMs show strong potential for classification, phase picking on traces above approximately 20 dB SNR, and magnitude estimation when forced to commit to a numeric output. The pipeline achieves a median location error of 9.6 km using LLM-derived P-picks (6.6 km after geometric quality control) and a mean magnitude absolute error of 0.295 magnitude units (median 0.125) on the requested magnitude scale, despite operating on raw-count waveforms without instrument-response correction. At the same time, performance is limited in tasks governed by station geometry, signal quality, and physical inversion constraints, and the polarity ablation in Section 4.5 demonstrates that first-motion polarity reading remains at chance even when the input window is aligned with the analyst’s published pick.

Future work should focus on extending the catalog to higher-SNR and larger-magnitude regimes where polarity reading may become tractable, integrating physics-informed constraints into the LLM prompt, evaluating performance at larger scales using compute clusters, and replacing the LLM polarity step with a deterministic signal-processing routine for downstream focal-mechanism estimation. A natural baseline-comparison line of work is to substitute the seismogram-interpretation pass’s P-pick output with picks from purpose-built deep-learning seismic pickers such as PhaseNet (30) or

EQTransformer (31) and re-run only the LLM-bottlenecked stages (magnitude, polarity), to isolate which parts of the catalog pipeline are constrained by general multimodal vision capability versus domain-specific signal processing. More broadly, these results suggest that multimodal LLMs may serve as useful assistants in seismological workflows when paired with deterministic algorithms that provide formal uncertainty diagnostics, and when their limitations are well understood and appropriately managed.

Code and Data Availability

The full catalog-pipeline code (the main notebook `earthquake_catalog.ipynb`, the helper module `catalog_normalization.py`, the polarity-comparison script, and the companion-site generator), all prompts, the generated CSV outputs that drive every quantitative claim in this paper (`master_summary.csv`, `step1_gemini.csv`, `step2_polarity.csv`, `usgs_pick_detail.csv`, `polarity_compare.csv`, `locations.csv`, `focal_mechanisms.csv`), and per-event station metadata are released as (1) at <https://github.com/AdvikarA/Seismo-LLM>. Waveform binaries are not committed and are re-fetched from FDSN data centers (15) on first run via the cache loop in the main notebook, so the pipeline reproduces from cold start with no manual data download.

References

- [1] A. Ananthkumar and M. Mousavi. *Seismo-LLM: End-to-End Earthquake Catalog Pipeline Driven by a Multimodal LLM*, version 1.0, 2026. <https://github.com/AdvikarA/Seismo-LLM>.
- [2] Gemini Team. *Gemini: A Family of Highly Capable Multimodal Models*. arXiv:2312.11805, 2023.
- [3] M. Reid et al. (Gemini Team). *Gemini 1.5: Unlocking Multimodal Understanding Across Millions of Tokens of Context*. arXiv:2403.05530, 2024.
- [4] OpenAI. *GPT-4V(ision) System Card*. Technical report, 2023.
- [5] J.-B. Alayrac et al. Flamingo: A Visual Language Model for Few-Shot Learning. *NeurIPS*, 2022.
- [6] H. Liu, C. Li, Q. Wu, and Y. J. Lee. Visual Instruction Tuning. *NeurIPS*, 2023.
- [7] S. Yin, C. Fu, S. Zhao, K. Li, X. Sun, T. Xu, and E. Chen. *A Survey on Multimodal Large Language Models*. arXiv:2306.13549, 2024.
- [8] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR*, 2023.
- [9] T. Schick et al. Toolformer: Language Models Can Teach Themselves to Use Tools. *NeurIPS*, 2023.
- [10] A. M. Bran, S. Cox, O. Schilter, C. Baldassari, A. D. White, and P. Schwaller. Augmenting Large Language Models with Chemistry Tools. *Nature Machine Intelligence*, 6:525–535, 2024.
- [11] C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune, and D. Ha. *The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery*. arXiv:2408.06292, 2024.
- [12] S. M. Mousavi, Y. Sheng, W. Zhu, and G. C. Beroza. Stanford EArthquake Dataset (STEAD): A Global Data Set of Seismic Signals for AI. *IEEE Access*, 7:179464–179476, 2019.
- [13] J. Woollam, A. Rietbrock, A. Bueno, and S. De Angelis. SeisBench — A Toolbox for Machine Learning in Seismology. *Seismological Research Letters*, 93(3):1695–1709, 2022.
- [14] L. Krischer et al. ObsPy: A Bridge for Seismology into the Scientific Python Ecosystem. *Computational Science & Discovery*, 8:014003, 2015.
- [15] International Federation of Digital Seismograph Networks. *FDSN Web Service Specifications*. Technical specification, 2013.
- [16] D. Schorlemmer et al. QuakeML: Status of the XML-Based Seismological Data Exchange Format. *Annals of Geophysics*, 54(1):59–65, 2011.
- [17] H. P. Crotwell, T. J. Owens, and J. Ritsema. The TauP Toolkit: Flexible Seismic Travel-Time and Raypath Utilities. *Seismological Research Letters*, 70(2):154–160, 1999.

- [18] B. L. N. Kennett and E. R. Engdahl. Traveltimes for Global Earthquake Location and Phase Identification. *Geophysical Journal International*, 105(2):429–465, 1991.
- [19] L. Geiger. Herdbestimmung bei Erdbeben aus den Ankunftszeiten. *Nachrichten von der Königlichen Gesellschaft der Wissenschaften zu Göttingen, Mathematisch-physikalische Klasse*, 331–349, 1910.
- [20] W. H. K. Lee and S. W. Stewart. *Principles and Applications of Microearthquake Networks*. Academic Press, 1981.
- [21] D. W. Marquardt. An Algorithm for Least-Squares Estimation of Nonlinear Parameters. *Journal of the Society for Industrial and Applied Mathematics*, 11(2):431–441, 1963.
- [22] F. W. Klein. *User's Guide to HYPOINVERSE-2000, a Fortran Program to Solve for Earthquake Locations and Magnitudes*. USGS Open-File Report 02-171, 2002.
- [23] F. Waldhauser and W. L. Ellsworth. A Double-Difference Earthquake Location Algorithm: Method and Application to the Northern Hayward Fault, California. *Bulletin of the Seismological Society of America*, 90(6):1353–1368, 2000.
- [24] J. L. Hardebeck and P. M. Shearer. A New Method for Determining First-Motion Focal Mechanisms. *Bulletin of the Seismological Society of America*, 92(6):2264–2276, 2002.
- [25] J. L. Hardebeck and P. M. Shearer. Using S/P Amplitude Ratios to Constrain the Focal Mechanisms of Small Earthquakes. *Bulletin of the Seismological Society of America*, 93(6):2434–2444, 2003.
- [26] G. Ekström, M. Nettles, and A. M. Dziewoński. The Global CMT Project 2004–2010: Centroid-Moment Tensors for 13,017 Earthquakes. *Physics of the Earth and Planetary Interiors*, 200–201:1–9, 2012.
- [27] L. K. Hutton and D. M. Boore. The M_L Scale in Southern California. *Bulletin of the Seismological Society of America*, 77(6):2074–2094, 1987.
- [28] H. Kanamori. The Energy Release in Great Earthquakes. *Journal of Geophysical Research*, 82(20):2981–2987, 1977.
- [29] T. C. Hanks and H. Kanamori. A Moment Magnitude Scale. *Journal of Geophysical Research: Solid Earth*, 84(B5):2348–2350, 1979.
- [30] W. Zhu and G. C. Beroza. PhaseNet: A Deep-Neural-Network-Based Seismic Arrival-Time Picking Method. *Geophysical Journal International*, 216(1):261–273, 2019.
- [31] S. M. Mousavi, W. L. Ellsworth, W. Zhu, L. Y. Chuang, and G. C. Beroza. Earthquake Transformer: An Attentive Deep-Learning Model for Simultaneous Earthquake Detection and Phase Picking. *Nature Communications*, 11:3952, 2020.